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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

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0718395175

MATHEMATICS (SYLLABUS D) **4024/22**

Paper 2 **October/November 2011**

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments
Electronic calculator

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.

Section B
Answer any **four** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.
You are expected to use an electronic calculator to evaluate explicit numerical expressions.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π, use either your calculator value or 3.142, unless the question requires the answer in terms of π.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 100.

For Examiner's Use

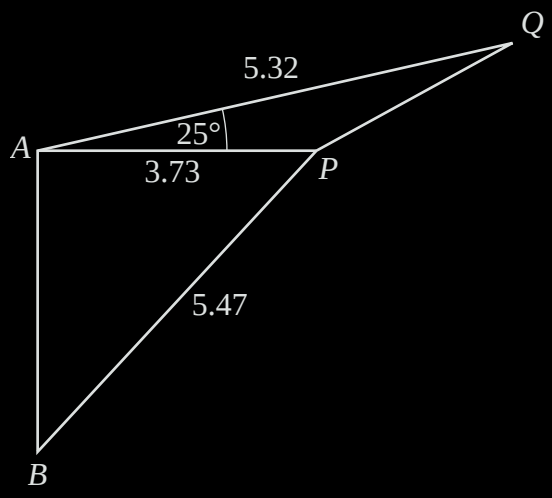
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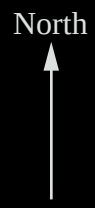
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The diagram shows four points, *A*, *B*, *P* and *Q*, at sea.
B is due South of *A* and *P* is due East of *A*.
AP = 3.73 km, *BP* = 5.47 km, *AQ* = 5.32 km and $P\hat{A}Q = 25°$.

(a) Calculate $A\hat{B}P$.

Answer .....[2]

(b) Calculate *PQ*.

Answer .....km [4]

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Diagram I

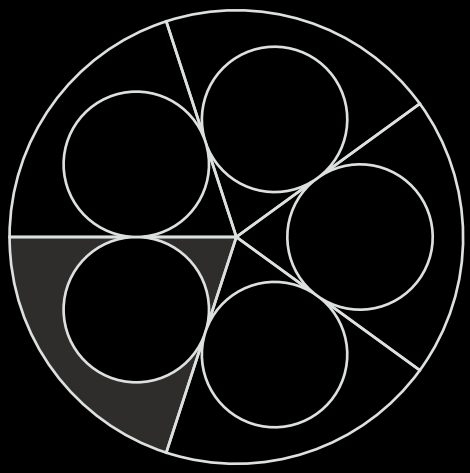


Diagram I shows one large circle and five identical small circles.
Each of the five radii shown is a tangent to two of the small circles.

(i) Describe the symmetry of the diagram.

Answer .....[1]

(ii) The radius of the large circle is R centimetres and the radius of each small circle is r centimetres.
Each small circle is equal in area to the shaded region.

Find $R^2 : r^2$.

Answer ..... : .....[3]

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Diagram II

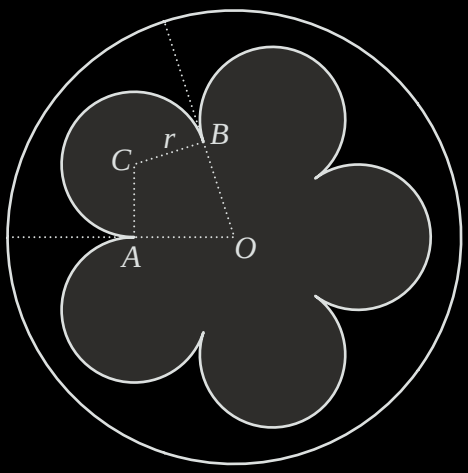


Diagram II shows the same large circle and arcs of the same small circles as in **Diagram I**.
C is the centre of one of the small circles.
This circle touches the adjacent circles at *A* and *B*.
O is the centre of the large circle.

(i) Show that reflex $A\hat{C}B = 252°$.

[2]

(ii) The perimeter of the shaded region is $k\pi r$ centimetres.

Calculate the value of *k*.

Answer ..... [2]

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$$\begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 11 \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}.$$

Find x and y.

Answer $x =$ .....

$y =$ ..... [2]

(b) **(i)** The transformation A is represented by the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Find, in terms of a, b, c and d as appropriate,

(a) the image of (1, 0) under the transformation A,

Answer (..... , ...) [1]

(b) the image of (0, 1) under the transformation A.

Answer (..... , ...) [1]

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[1]

(b) Find the gradient of AB.

Answer ..... [1]

(c) The equation of the line AB is $x + Py + Q = 0$.

Find P and Q.

Answer $P =$ .....

$Q =$ ..... [2]

(d) The point C is (12, 2).

(i) Given that C is the midpoint of BM, find the coordinates of M.

Answer (..... , ...) [1]

(ii) Calculate AC.

Answer ..... units [1]

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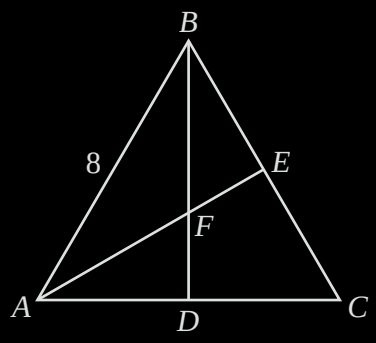


Diagram I

In **Diagram I**, *ABC* is an equilateral triangle of side 8 cm.
D and *E* are the midpoints of *AC* and *BC* respectively.
BD and *AE* intersect at *F*.

(a) (i) Find the area of triangle *ABC*.

Answer .....cm^2 [2]

(ii) Show that $A\hat{F}B = 120°$.

Answer .....

.....[1]

(iii) Calculate *AF*.

Answer ..... cm [2]

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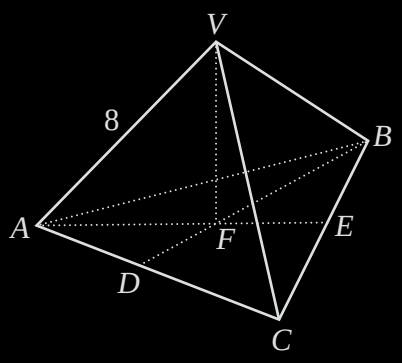


Diagram II

The equilateral triangle of side 8 cm in **Diagram I** forms the base of the triangular pyramid *VABC* in **Diagram II**.
The vertex *V* is vertically above *F*.
VA = *VB* = *VC* = 8 cm.

(i) Calculate the surface area of the pyramid.

Answer .....cm^2 [1]

(ii) Calculate the volume of the pyramid.

Answer .....cm^3 [3]

(c) A pyramid *P* is geometrically similar to *VABC* and its volume is $\frac{1}{64}$ of the volume of *VABC*.

(i) Find the length of an edge of *P*.

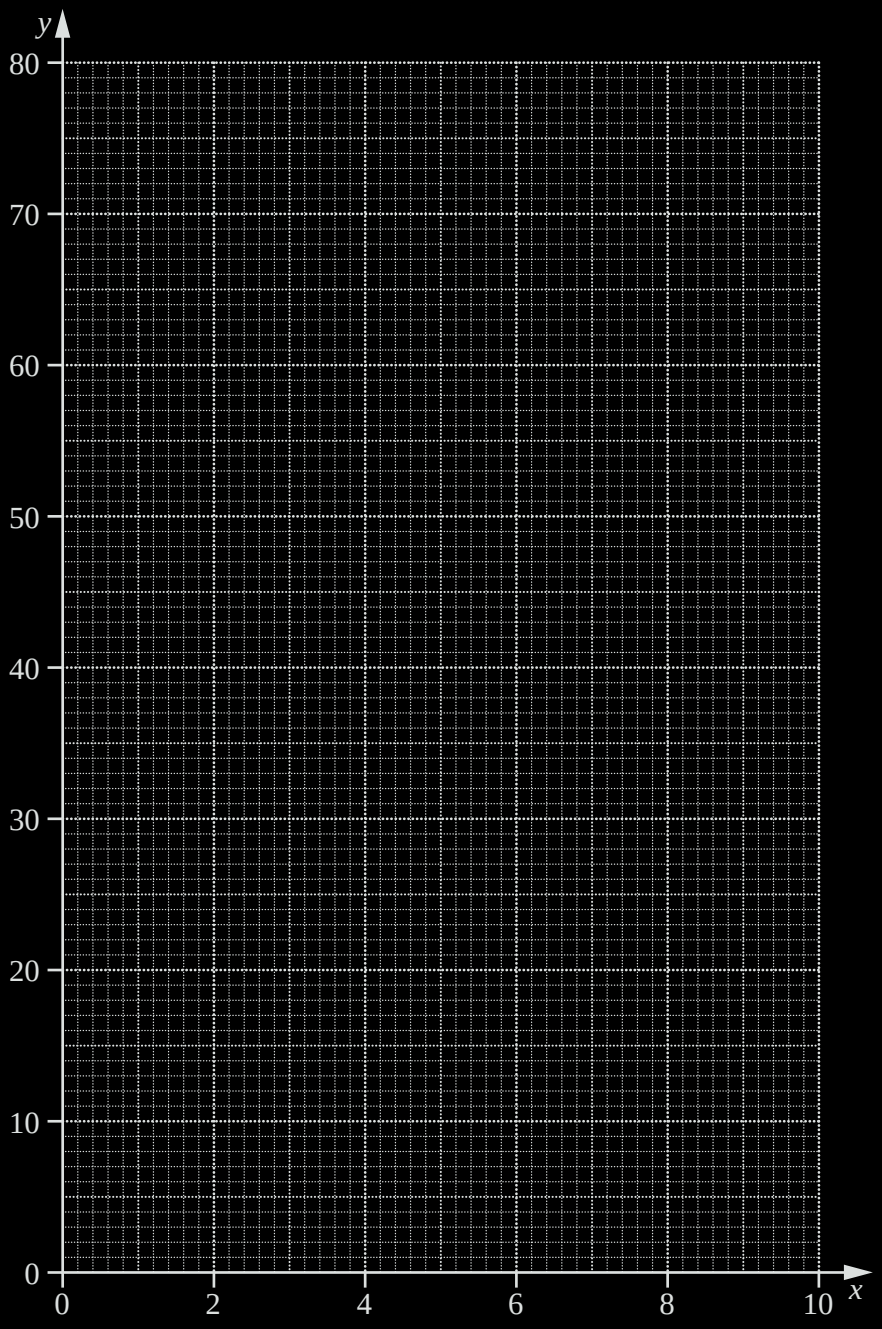
Answer ..... cm [2]

(ii) A pyramid that is identical to *P* is removed from each of the four vertices of *VABC*.

State the number of faces of the new solid.

Answer ..... [1]

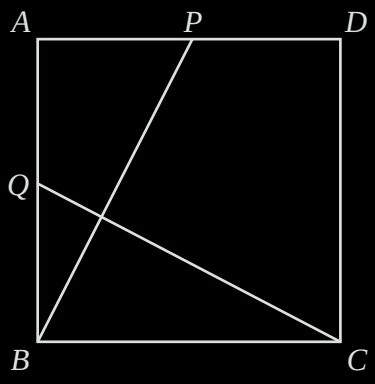
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Diagram I



In **Diagram I**, *ABCD* is a square.
P and *Q* are the midpoints of *AD* and *AB* respectively.

(a) Show that triangles *APB* and *BQC* are congruent.

Answer .....

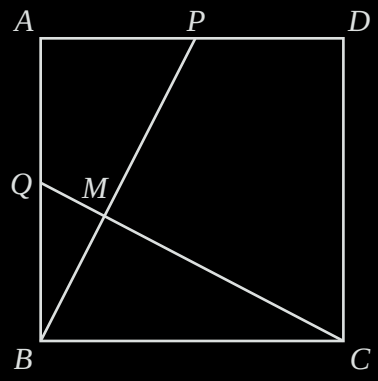
.....

.....

..... [3]

(b)

Diagram II



In **Diagram II**, *QC* and *PB* intersect at *M*.

Show that $B\hat{M}C = 90°$.
State your reasons clearly.

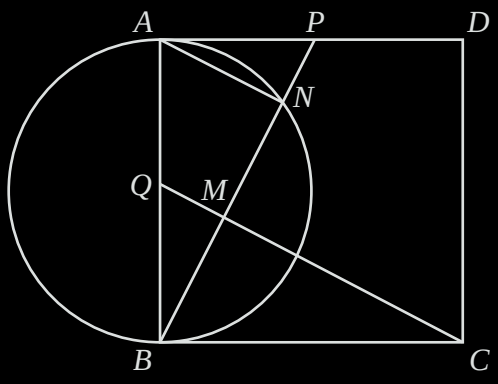
Answer .....

.....

..... [2]

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Diagram III



In **Diagram III**, the circle centre Q has diameter AB.
The circle intersects BP at N.

(i) State the reason why $A\hat{N}B = 90°$.

Answer .....[1]

(ii) Triangle BMQ is mapped onto triangle BNA by an enlargement.
Write down the centre and scale factor of the enlargement.

Answer .....

..... [1]

(iii) Given that $QM = 3$ cm,

(a) find AN,

Answer ..... cm [1]

(b) show that $MN = 6$ cm,

Answer .....

.....[1]

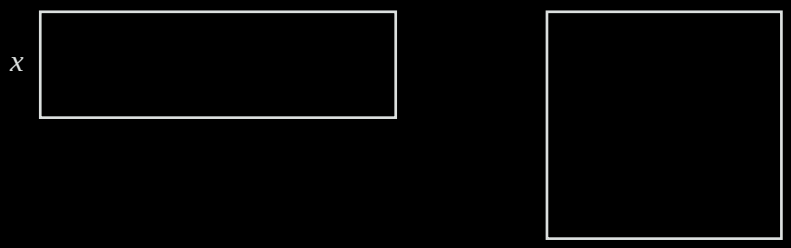
(c) find MC,

Answer ..... cm [1]

(d) find the area of triangle APB.

Answer .....cm^2 [2]

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A piece of wire, 28 cm in length, is cut into two parts.
One part is used to make a rectangle and the other a square.

The length of the rectangle is three times its width.
The width of the rectangle is x centimetres.

(a) (i) Write down an expression, in terms of x, for the length of the rectangle.

Answer ..... cm [1]

(ii) Find, and simplify, an expression, in terms of x, for the length of a side of the square.

Answer ..... cm [2]

(b) It is given that the area of the rectangle is equal to the area of the square.

(i) Form an equation in x and show that it reduces to $x^2 - 28x + 49 = 0$.

[2]

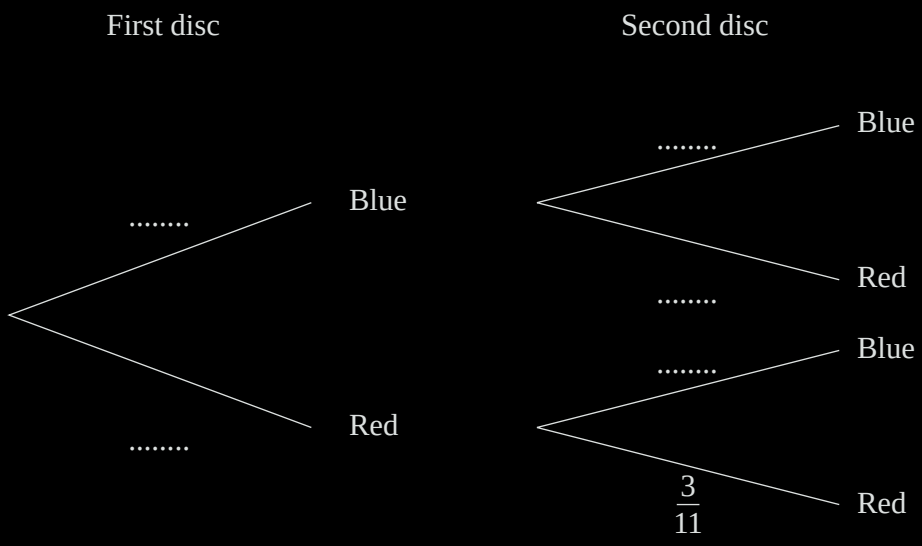
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(i) Complete the tree diagram. [2]

(ii) Expressing each of your answers as a fraction **in its lowest terms**, calculate the probability that

(a) both discs are red,

Answer ..... [1]

(b) at least one disc is blue.

Answer ..... [2]

(iii) A third disc is picked out at random.
Calculate the probability that all three discs are red.

Answer ..... [1]

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